Continuum Mechanics
Class # 16
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Chapter 5: Balance Laws

Mass and Momentum Conservation

Definition: Let $\vec{\phi} : B \times [0,\infty) \to \mathbb{E}^3$ be a motion. Set

$$B_t = \vec{\phi}_t(B).$$

Let $\rho(\cdot,\cdot) : B_t \times [0,\infty) \to \mathbb{R}$ denote the mass density of a material and suppose $\vec{v}(\vec{x},t)$ is the spatial velocity field. Let $\Omega_t \subseteq B_t$ be arbitrary. The mass of Ω_t is

$$\text{mass}[\Omega_t] := \int_{\Omega_t} \rho(\vec{x},t)\, dV_x.$$

The linear momentum of Ω_t is

$$\vec{l}[\Omega_t] := \int_{\Omega_t} \rho(\vec{x},t)\vec{v}(\vec{x},t)\, dV_x$$

and the angular momentum of Ω_t wrt $\vec{z} \in \mathbb{E}^3$ is

$$\vec{J}[\Omega_t]_{\vec{z}} := \int_{\Omega} (\vec{x}-\vec{z}) \times [\rho(\vec{x},t)\vec{v}(\vec{x},t)]\, dV_x. \quad ///$$

Axiom (5.1): In the absence of chemical reactions and neglecting relativistic effects, the mass of a material undergoing motion $\vec{\phi}$ in the set Ω_t does not change over time, that is,

$$\frac{d}{dt}\,\text{mass}[\Omega_t] = 0\,, \quad \forall\, \Omega_t \subseteq B_t\,,$$

for all $t \in [0,\infty)$. ///

Definition: Suppose that ϕ is a motion. Suppose that $\Omega_t \subseteq B_t = \phi_t(B)$ is subjected to the body force

$$\vec{b} : B_t \times [0,\infty) \to \mathcal{V}$$

and external surface traction

$$\vec{t} : \partial\Omega_t \times [0,\infty) \to \mathcal{V}\,.$$

The resultant force on Ω_t is

$$\vec{r}[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t)\,\vec{b}(\vec{x},t)\,dV_x + \int_{\partial\Omega_t} \vec{t}(\vec{x},t)\,dA_x$$

and the resultant torque on Ω_t about $\vec{z} \in \mathbb{E}^3$ is

$$\vec{\tau}[\Omega_t]\vec{z} = \int_{\Omega_t} (\vec{x}-\vec{z}) \times \rho(\vec{x},t)\vec{b}(\vec{x},t)\,dV_x$$

$$+ \int_{\partial\Omega_t} (\vec{x}-\vec{z}) \times \vec{t}(\vec{x},t)\,dA_x\,. \quad ///$$

Axiom (5.2): With respect to a fixed frame of reference, the rate of change of linear momentum of any $\Omega_t \subseteq B_t$ equals the resultant force applied to it.

The rate of change of angular momentum about the origin equals the the resultant torque about the origin. In symbols,

(5.7)
$$\frac{d}{dt}\vec{L}[\Omega_t] = \mathcal{T}[\Omega_t], \qquad \forall\, \Omega_t \subseteq B_t.$$
$$\frac{d}{dt}\vec{J}[\Omega_t]_{\vec{0}} = \vec{\mathcal{T}}[\Omega_t]_{\vec{0}}, \quad \text{///}$$

Remark: These balances come from the study of discrete particle systems where any pair of particles interact with an energy and every particle is acted upon by an external force.

One can view a continuum system as the limit of discrete particle systems.

The book has a nice discussion of particle systems in Section 5.1. ///

The First Law of Thermodynamics

Definition: Let $\vec{\phi}$ be the motion of a continuum body with mass density ρ and spatial velocity $\vec{v}$. Let $\Omega_t \subseteq B_t = \vec{\phi}_t(B)$ be arbitrary, the internal energy of Ω_t is defined as

$$U[\Omega_t] := \int_{\Omega_t} \rho(\vec{x},t)\, \phi(\vec{x},t)\, dV_x$$

where $\phi : B_t \times [0,\infty) \to \mathbb{R}$ is the internal energy density per unit mass or specific internal energy

The quantity $Q[\Omega_t]$ is called the net heating of Ω_t and is defined as

$$Q_t[\Omega_t] := \int_{\Omega_t} \rho(\vec{x},t)\, r(\vec{x},t)\, dV_x - \int_{\partial\Omega_t} \vec{q}(\vec{x},t)\cdot\vec{n}(\vec{x})\, dA_x \tag{5.13}$$

where $r : B_t \times [0,\infty) \to \mathbb{R}$ is the heat supply per unit mass or specific heat supply and $\vec{q} : B_t \times [0,\infty) \to \mathbb{R}$ is the Fourier-Stokes heat flux. ///

Definition: Let $\vec{\phi}$ be the motion of a continuum body with mass density ρ and spatial velocity $\vec{v}$. Let $\Omega_t \subseteq B_t = \vec{\phi}_t(B)$ be arbitrary. Assume that $\vec{b}$ is a spatial body force and $\vec{t}$ is a surface traction on $\partial\Omega_t$. The power of external forces is defined as

$$P[\Omega_t] := \int_{\Omega_t} \rho(\vec{x},t)\vec{b}(\vec{x},t)\cdot\vec{v}(\vec{x},t)\, dV_x + \int_{\partial\Omega_t} \vec{t}(\vec{x},t)\cdot\vec{v}(\vec{x},t)\, dA_x$$

The kinetic energy of Ω_t is

$$K[\Omega_t] := \int_{\Omega_t} \frac{1}{2}\rho(\vec{x},t)\,|\vec{v}(\vec{x},t)|^2\, dV_x .$$

The net working of Ω_t by external forces is defined as

$$W[\Omega_t] := P[\Omega_t] - \frac{d}{dt} K[\Omega_t]. \quad (5.14)$$ ///

Remark: If $W[\Omega_t] = 0$, for $t \in [t_0, t_1]$ then we say that all of the applied mechanical energy delivered to a continuum body over $[t_0, t_1]$ is used to produce or change the motion. ///

Axiom (5.3) (The First Law of Thermodynamics) Considering only thermal and mechanical energy, the rate of change of internal energy of Ω_t equals the sum of net heating and net working applied to Ω_t, that is,

$$\frac{d}{dt} U[\Omega_t] = Q[\Omega_t] + W[\Omega_t]$$

for any $\Omega_t \subseteq B_t$. Equivalently, using (5.14) we have

$$\frac{d}{dt}[U[\Omega_t] + K[\Omega_t]] = Q[\Omega_t] + P[\Omega_t]. \quad ///$$

Proposition: Suppose that the power of external forces can be expressed as

$$P[\Omega_t] = -\frac{d}{dt} G[\Omega_t],$$

where $G[\Omega_t]$ is called the potential energy

for external forces. Then the First Law is equivalent to

$$\frac{d}{dt}\left(U[\Omega_t] + K[\Omega_t] + G[\Omega_t]\right) = Q[\Omega_t]. \quad ///$$

Entropy and The Second Law of Thermodynamics

Axiom (5.4): (The Second Law of Thermodynamics) There exists a quantity called entropy defined as

$$H[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t)\,\eta(\vec{x},t)\,dV,$$

where η is the entropy density per unit mass or specific entropy. There exists a temperature field denoted $\theta(\vec{x},t)$ which is everywhere positive. The time rate of change of entropy minus the heating per unit temperature is always non-negative, that is

$$\frac{d}{dt}H[\Omega_t] - \left[\int_{\Omega_t} \frac{\rho(\vec{x},t)\,r(\vec{x},t)}{\theta(\vec{x},t)}\,dV_x - \int_{\partial\Omega_t} \frac{\vec{q}(\vec{x},t)\cdot\vec{n}(\vec{x})}{\theta(\vec{x},t)}\,dA_x\right]$$

$$= \int_{\Omega_t} \sigma(\vec{x},t)\,dV_x$$

$$\geq 0$$

for all $\Omega_t \subseteq B_t$. $\sigma(\vec{x},t)$ is called the entropy production rate. ///

Remark: In the absence of any heating of Ω_t, we have

$$\frac{d}{dt}\mathcal{H}[\Omega_t] = \int_{\Omega} \sigma(\vec{x},t)\, dV_x \geq 0.$$

When no heating/cooling is applied the entropy in Ω_t naturally increases.

Because of the arbitrariness of Ω_t, we can show that $\sigma(\vec{x},t) \geq 0 \quad \forall\ \vec{x} \in B_t,\ t \geq 0$.

The inequality in Axiom (5.4) is called the Clausius-Duhem Inequality.

The Continuity Equation

Result (5.5): (Conservation of Mass) Let $\vec{\phi}: B \times [0,\infty) \to \mathbb{E}^3$ be a motion of a continuum body with associated spatial velocity field $\vec{v}(\vec{x},t)$ and spatial mass field $\rho(\vec{x},t)$. Then Axiom (5.1) is equivalent to

$$\frac{\partial}{\partial t}\rho + \nabla^x \cdot (\rho \vec{v}) = 0, \quad \forall\ \vec{x} \in B_t,\ t \geq 0,$$

or, equivalently,

$$\dot{\rho} + \rho \nabla^x \cdot \vec{v} = 0, \quad \forall\ \vec{x} \in B_t,\ t \geq 0.$$

Proof: ($\Longrightarrow$) Assume that Axiom (5.1) holds. Then

$$\text{mass}[\Omega_t] = \text{mass}[\Omega_0] \qquad \forall\ \Omega_t \subseteq B_t$$

for all $t \geq 0$. By Result (4.8),

$$\text{mass}[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t)\, dV_x = \int_{\Omega} \rho_m(\vec{X},t)\, J(\vec{X},t)\, dV_X$$

where ρ_m is the material description of the spatial mass density ρ, ie,

$$\rho_m(\vec{X},t) = \rho(\vec{\varphi}(X,t),t).$$

Since $\vec{\varphi}(\vec{X},0) = \vec{X}$,

$$\text{mass}[\Omega_0] = \int_{\Omega} \rho(\vec{X},0)\, dV_X$$

$$= \int_{\Omega} \rho_0(\vec{X})\, dV_X,$$

where

$$\rho_0(\vec{X}) := \rho(\vec{X},0).$$

The integral form of mass conservation implies

$$\int_{\Omega} \left[\rho_m(X,t)\, J(\vec{X},t) - \rho_0(X)\right] dV_X = 0,$$

for any $\Omega \subseteq B$ and any $t \geq 0$. By the Localization Theorem

(5.18)
$$\rho_m(\vec{X},t)\, J(\vec{X},t) = \rho_0(\vec{X}), \quad \forall\, \vec{X} \in B, \ \forall\, t \geq 0.$$

This is a material (or Lagrangian) form of local conservation. We want a spatial (or Eulerian) form.

Taking the time derivative of (5.18), we have

$$\frac{\partial}{\partial t}\rho_m(\vec{X},t)\,J(\vec{X},t)+\rho_m(\vec{X},t)\frac{\partial}{\partial t}J(\vec{X},t)=0.$$

Using Result (4.9),

$$\frac{\partial}{\partial t}\rho_m(\vec{X},t)J(\vec{X},t)+\rho_m(\vec{X},t)J(\vec{X},t)\left[\nabla^x\cdot\vec{v}\right]_m(\vec{X},t)=0$$

Since $J(\vec{X},t)>0$, for all admissible motions,

$$\frac{\partial}{\partial t}\rho_m(X,t)+\rho_m(\vec{X},t)\left[\nabla^x\cdot\vec{v}\right]_m(\vec{X},t)=0$$

Converting back to a spatial (Eulerian) description,

$$\dot{\rho}(\vec{x},t)+\rho(\vec{x},t)\,\nabla^x\cdot\vec{v}(\vec{x},t)=0.$$

Using Result (4.7)

$$\frac{\partial}{\partial t}\rho(\vec{x},t)+\nabla^x\rho(\vec{x},t)\cdot\vec{v}(\vec{x},t)+\rho(\vec{x},t)\,\nabla^x\cdot\vec{v}(\vec{x},t)=0$$

which may be written

$$\frac{\partial}{\partial t}\rho(\vec{x},t)+\nabla^x\cdot\left(\rho(\vec{x},t)\vec{v}(\vec{x},t)\right)=0.$$

($\Leftarrow$) This direction is an exercise. //

Remark: We will need the material conservation equation (5.18) in a later calculation. Had we no need of that there is a two line proof we could have used for the forward direction. //

Shorter Proof. ($\Longrightarrow$) Axiom 5.1 may be written

$$0 = \frac{d}{dt}\int_{\Omega_t} \rho(\vec{x},t)\, dV_x$$

Using RTT,

$$\frac{d}{dt}\int_{\Omega_t} \rho(\vec{x},t)\, dV_x = \int_{\Omega_t} \left\{ \dot{\rho} + \rho(\nabla^x \cdot \vec{v}) \right. dV_x$$

$$= \int_{\Omega_t} \left\{ \frac{\partial \rho}{\partial t} + \nabla^x \rho \cdot \vec{v} + \rho \nabla^x \cdot \vec{v} \right\} dV_x$$

$$= \int_{\Omega_t} \left\{ \frac{\partial \rho}{\partial t} + \nabla^x \cdot (\rho \vec{v}) \right\} dV_x .$$

Using the localization Theorem, we obtain

$$\frac{\partial \rho}{\partial t} + \nabla^x \cdot (\rho \vec{v}) = 0, \quad \forall\, \vec{x} \in B_t, \forall\, t \geq 0. \quad ///$$

This last equation is called the <u>continuity equation</u>.

<u>Result</u> (5.6): Let $\vec{\phi} : B \times [0,\infty) \to \mathbb{E}^3$ be a motion with associated spatial velocity field $\vec{v}(\vec{x},t)$ and mass density $\rho(\vec{x},t)$. Let q be any spatial scalar field. Suppose $\Omega_t \subseteq B_t = \vec{\phi}_t(B)$ be arbitrary. Then

$$\frac{d}{dt}\int_{\Omega_t} q(\vec{x},t)\rho(\vec{x},t)\, dV_x = \int_{\Omega_t} \dot{q}(\vec{x},t)\rho(\vec{x},t)\, dV_x$$

Proof: By the Reynolds Transport Theorem

$$\frac{d}{dt}\int_{\Omega_t} q(\vec{x},t)\,\rho(\vec{x},t)\,dV_x$$

$$= \int_{\Omega_t}\left(\frac{\partial}{\partial t}(q\rho) + \nabla^x(q\rho)\cdot\vec{v} + q\rho\,(\nabla^x\cdot\vec{v})\right) dV_x$$

$$= \int_{\Omega_t}\Big(\frac{\partial q}{\partial t}\rho + q\frac{\partial \rho}{\partial t} + \nabla^x q\cdot(\rho\vec{v}) + \nabla\rho\cdot(q\vec{v}) + q\rho\,(\nabla^x\cdot\vec{v})\Big)\,dV_x$$

$$= \int_{\Omega_t}\dot{q}\,\rho\,dV_x + \int_{\Omega_t} q\left(\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v})\right) dV_x$$

$$= \int_{\Omega_t}\dot{q}\,\rho\,dV_x\,,$$

using the continuity equation. ///

Remark: The book's proof of Result (5.6) is much more complicated. //

Local Conservation of Linear Momentum

Result (5.7): Let $\vec{\phi}: B \times [0,\infty) \to \mathbb{E}^3$ be a motion of a continuum body with associated spatial velocity field $\vec{v}(\vec{x},t)$ and mass density $\rho(\vec{x},t)$. Axiom (5.2) (balance of linear momentum) is equivalent to

$$\rho \dot{\vec{v}} = \nabla^x \cdot S + \rho \vec{b}, \quad \forall \vec{x} \in B_t, \; t \geq 0,$$

where $S(\vec{x},t)$ is the Cauchy stress field and $\vec{b}(\vec{x},t)$ is the spatial body force per unit mass.

Proof: ($\Longrightarrow$) Assume that Axiom (5.2) holds. Then

$$\frac{d}{dt} \int_{\Omega_t} \rho(\vec{x},t) \vec{v}(\vec{x},t) \, dV_x$$

$$= \int_{\partial\Omega_t} \vec{t}(\vec{x},t) \, dA_x + \int_{\Omega_t} \rho(\vec{x},t) \vec{b}(\vec{x},t) \, dV_x$$

$$= \int_{\partial\Omega_t} S\vec{n} \, dA_x + \int_{\Omega_t} \rho(\vec{x},t) \vec{b}(\vec{x},t) \, dV_x$$

$$= \int_{\Omega_t} \left(\nabla \cdot S + \rho(\vec{x},t) \vec{b}(\vec{x},t) \right) dV_x$$

using the Divergence Theorem and

$$\vec{t} = S\vec{n} \quad \text{on} \quad \partial\Omega_t$$

Using Result (5.6)

$$\frac{d}{dt}\int_{\Omega_t} \rho\, \vec{v}\, dV_x = \int_{\Omega_t} \rho\, \dot{\vec{v}}\, dV_x$$

The result now follows from the Localization Theorem. ///